Centrifugal Process Pump (PUMP-CENT) Equipment Manual and SOP

VINDHYA PRECISION WORKS
STANDARD OPERATING PROCEDURE & EQUIPMENT MANUAL
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Asset Class: PUMP-CENT (Centrifugal Process Pump)
Effective Date: 14 January 2026
Maintenance Discipline: Mechanical (Primary), Instrumentation (Support)

1.0 ASSET OVERVIEW AND SCOPE

1.1 This document defines the operating limits, preventive maintenance schedules, fault diagnosis protocols, torque specifications, and safety-critical isolation procedures for all Centrifugal Process Pumps (Asset Class: PUMP-CENT) installed across Vindhya Precision Works.

1.2 The parameters, limits, and part numbers contained in this manual are specific to asset class PUMP-CENT and shall not be applied to positive displacement pumps, screw compressors, or auxiliary blowers.

2.0 NORMAL OPERATING RANGES AND CONDITION MONITORING SETPOINTS

2.1 Healthy units within asset class PUMP-CENT operate within the baseline parameters defined in Table 2.1.

Table 2.1: Baseline Operating Ranges and Alarm Setpoints
Parameter	Healthy Operating Range	Warning Setpoint	Trip Setpoint
Vibration (RMS Velocity)	$1.2\text{--}4.0\text{ mm/s}$	5.0 mm/s	8.0 mm/s
Surface Temperature	$38\text{--}66\ ^\circ\text{C}$	$75\ ^\circ\text{C}$	$90\ ^\circ\text{C}$
Operating Pressure	4.2 bar (nominal)	3.4 bar (Low-Pressure Alarm)	N/A
Motor Current	$28.0\text{--}34.5\text{ Amps}$	38.0 Amps	42.0 Amps

2.2 Any reading exceeding the Warning Setpoint shall be logged by the condition-monitoring system and reported to the Mechanical Maintenance Lead within two (2) operating shifts.

2.3 Any reading reaching the Trip Setpoint shall automatically de-energize the pump motor via the Plant Distributed Control System (DCS) within 500 milliseconds and trigger a high-priority technician dispatch.

3.0 PREVENTIVE MAINTENANCE (PM) SCHEDULE

3.1 The base preventive maintenance interval for asset class PUMP-CENT is exactly 4000 running hours, or every 180 calendar days, whichever occurs first.

3.2 PM-1 (Every 1000 Running Hours): Inspect mechanical seal flush lines for obstruction; record motor current and bearing housing vibration signatures manually; verify lubrication levels in the bearing housings.

3.3 PM-2 (Every 4000 Running Hours - Minor Overhaul): Execute full machinery shutdown, apply mandatory lock-out/tag-out (LOTO), replace VPW-P-00007 (Rate filter) and VPW-P-00023 (Discussion filter), inspect impeller clearance, and test trip interlocks.

3.4 PM-3 (Every 12000 Running Hours - Major Overhaul): Complete disassembly of pump wet-end; replace rotating assembly bearings using VPW-P-00020 (Once bearing) and VPW-P-00031 (Western bearing); replace shaft sealing elements using VPW-P-00026 (Second seal) and VPW-P-00030 (Author seal); recalibrate instrumentation via VPW-P-00011 (Nation sensor).

4.0 FAULT CODES AND DIAGNOSTIC PROSE

4.1 Fault Code FC-PC-101 (High Vibration): Probable causes include pump-motor shaft misalignment exceeding 0.05 mm, cavitation due to inlet starvation, or bearing raceway spalling. Corrective action: Check laser alignment of coupling; verify suction pressure is $\ge 3.4\text{ bar}$; if vibration persists above 8.0 mm/s trip threshold after alignment, replace bearings VPW-P-00020 and VPW-P-00031.

4.2 Fault Code FC-PC-102 (High Surface Temperature): Probable causes include dry running, degraded bearing grease, or external friction from the mechanical seal faces. Corrective action: Stop pump immediately if temperature reaches $90\ ^\circ\text{C}$; inspect seal flush line fitted with VPW-P-00023 (Discussion filter) for blockages; clean or replace filter element; repack bearing cavities with high-temperature synthetic grease.

4.3 Fault Code FC-PC-103 (Low Operating Pressure): Probable causes include worn impeller wear rings, air entrainment in the suction header, or a partially closed suction isolation valve (VPW-P-00033). Corrective action: Verify valve position on VPW-P-00033; bleed air from casing vent port; inspect internal hydraulic clearances; replace worn wear rings if differential head drops below 3.1 bar at rated speed.

4.4 Fault Code FC-PC-104 (Motor Overcurrent): Probable causes include process fluid specific gravity exceeding design limits ($\text{SG} > 1.15$), mechanical rub between the impeller and casing, or motor winding phase imbalance. Corrective action: Measure phase-to-phase electrical resistance; inspect pump wet-end for foreign debris; verify process fluid density via laboratory sampling.

4.5 Fault Code FC-PC-105 (Seal Leakage / Flush Failure): Probable causes include particulate scoring of seal faces or restriction in the internal flush circuit. Corrective action: Isolate hydraulic energy; remove and inspect mechanical seal assembly; replace secondary seal components VPW-P-00026 and VPW-P-00030; inspect and replace inline particulate filter VPW-P-00007.

4.6 Fault Code FC-PC-106 (Instrumentation Drift / Fault): Probable causes include electronic calibration drift, signal loop degradation, or sensor tip fouling. Corrective action: Instrument technician to cross-check local gauge readings against DCS input; recalibrate or replace faulty sensor element VPW-P-00011 (Nation sensor) in accordance with calibration SOP VPW-SOP-INS-019.

5.0 CONSUMABLE AND SPARE PARTS LIST

5.1 Maintenance technicians shall strictly use the plant inventory part numbers listed in Table 5.1 for all maintenance and overhaul activities on asset class PUMP-CENT. No substitutions are permitted without prior written authorization from the Engineering Manager.

Table 5.1: Inventory Spare Parts and Consumables

Inventory Part Number	Part Description
VPW-P-00007	Rate filter
VPW-P-00011	Nation sensor
VPW-P-00020	Once bearing
VPW-P-00023	Discussion filter
VPW-P-00026	Second seal
VPW-P-00030	Author seal
VPW-P-00031	Western bearing
VPW-P-00033	Trade valve

6.0 TORQUE SPECIFICATIONS

6.1 All mechanical fasteners associated with asset class PUMP-CENT shall be tightened using calibrated torque wrenches in a star-pattern sequence to the specific limits set forth in Table 6.1.

Table 6.1: Fastener Torque Limits

Fastener Location	Thread Size	Target Torque (Nm)
Casing Bolt / Studs	M16	$115\text{ Nm} \pm 5\%$
Bearing Housing Cap Screws	M10	$45\text{ Nm} \pm 5\%$
Coupling Hub Clamp Bolts	M12	$78\text{ Nm} \pm 5\%$
Seal Gland Nuts	M8	$22\text{ Nm} \pm 5\%$
Baseplate Anchor Bolts	M20	$180\text{ Nm} \pm 10\%$

7.0 HAZARDOUS ENERGY ISOLATION AND SAFETY-CRITICAL CORRECTIVE ACTIONS

7.1 Hazardous Energy Sources: Prior to initiating any invasive maintenance, disassembly, or corrective work on asset class PUMP-CENT, technicians must isolate two (2) primary hazardous energy sources:
1. Electrical Energy: 415V AC 3-phase supply via motor control center (MCC) breaker rack-out and padlocked tag-out.
2. Hydraulic Energy: Process fluid pressure via positive closure of suction and discharge isolation valves (including Trade valve VPW-P-00033), followed by atmospheric venting and draining of the pump casing.

7.2 Safety-Critical Corrective Actions: The following corrective actions are classified as safety-critical. They strictly mandate the execution of a formal Lockout/Tagout (LOTO) procedure under an active Permit to Work (PTW) issued by the Shift Safety Officer:
* Replacement or adjustment of mechanical seals (involving components VPW-P-00026 and VPW-P-00030) on pumps handling hazardous or corrosive process media.
* Disconnection of primary electrical power leads at the motor terminal box.
* Replacement of rotating element bearings (VPW-P-00020 and VPW-P-00031) requiring overhead crane lifting or heavy rigging.
* Opening of the pump casing or associated piping spool pieces where residual chemical or thermal energy may be trapped behind closed valves.

7.3 Non-invasive tasks—such as replacing exterior instrumentation filters (VPW-P-00007, VPW-P-00023), reading vibration signatures, or inspecting non-pressure-retaining safety guards—may be performed under a standard plant maintenance clearance without a formal PTW, provided running machinery is not contacted.